

Governing AI: introduction to the theories and principles of regulation

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Recap of last seminar

- AI risks (3 levels)
 - Privacy is in threat (micro level)
 - Algorithmic shadowing (micro level)
 - Jailbreaking, discrimination, disinformation and hallucination (output phase)
 - AI anthropomorphism
 - Enterprise and Jurisdictions reactions (meso level)
 - Science, Societal and Broader implications (macro level)
 - (...)
- Solutions
 - UNESCO: 'Recommendation on the Ethics of AI' (November 2021)
 - Soft regulation (ethics, standards and codes);
 - Hard regulation (legally binding laws and frameworks)
 - Combination of both soft and hard regulation

Outline

1. Theories of Regulation
2. AI Principles as normative frameworks
3. How theories and principles interact? Regulatory approaches to AI regulation
 - Examples of regulatory initiatives in the EU and UK
4. Group activity



1. Theories of Regulation

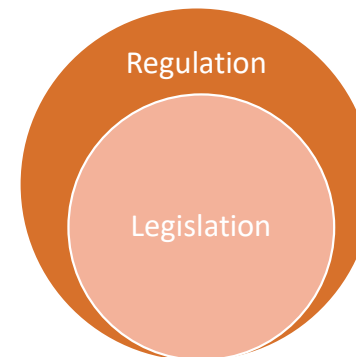
What is regulation?

- Legislation: a law developed by a legislative authority, such as a parliament or a congress. Legislation typically sets out a general set of principles, rules, and requirements that are required by law
- **Regulation**: a set of rules and requirements that are developed and enforced by a regulatory authority. Regulations will typically outline more specific activities, rules, and requirements than the general outline provided by legislation → **regulation as governance/oversight ecosystem**
- **Regulation** is a system of organisational practices and tools that collectively influence people to act in accordance with shared rules and norms
- Two approaches to thinking about the legislation vs. regulation distinction (not mutually exclusive):

Jurisprudence perspective
Legislation as a general outline for regulation

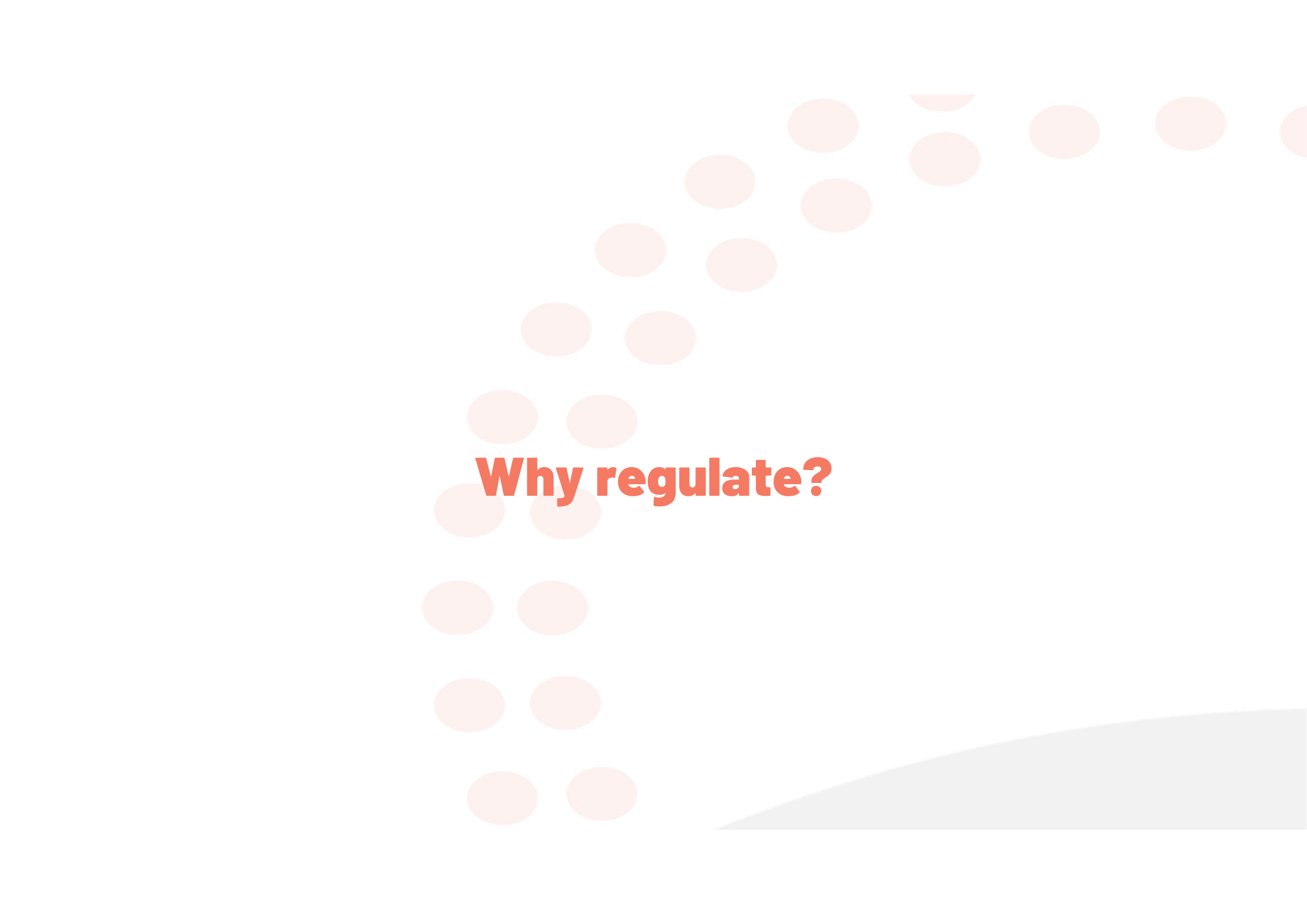


Regulatory governance perspective
Legislation as one of many regulatory tools



Theoretical foundations of AI regulation

- **AI regulation is complex:** AI impacts multiple sectors (healthcare, transport, finance, etc.), creating ethical, legal, and social challenges like bias, lack of accountability, and transparency.
- **Theories guide effective policy:** Understanding regulatory theories (e.g., public interest, private interest, institutionalist) helps us create balanced, adaptive policies that align AI development with societal values.
 - These theories provide a framework for tackling systemic risks and ensuring equity, fairness, and privacy in AI systems.
- **Learn from the past:** Theoretical framework is established on the experience. Learning from it can help navigate global and national regulatory approaches, drawing lessons e.g., from Internet regulation etc.

The background features a series of light pink circles of varying sizes arranged in a pattern that suggests movement or a trail, starting from the bottom left and moving towards the top right. A soft, grey, wave-like shape is positioned at the bottom right of the frame.

Why regulate?

Theories of regulation

Public interest theories (explain why governments regulate)

- Market failure rationales
- Rights-based and societal rationales
- Procedural approach

Why governments regulate? (Public interest)

Market failure rationale (economic):

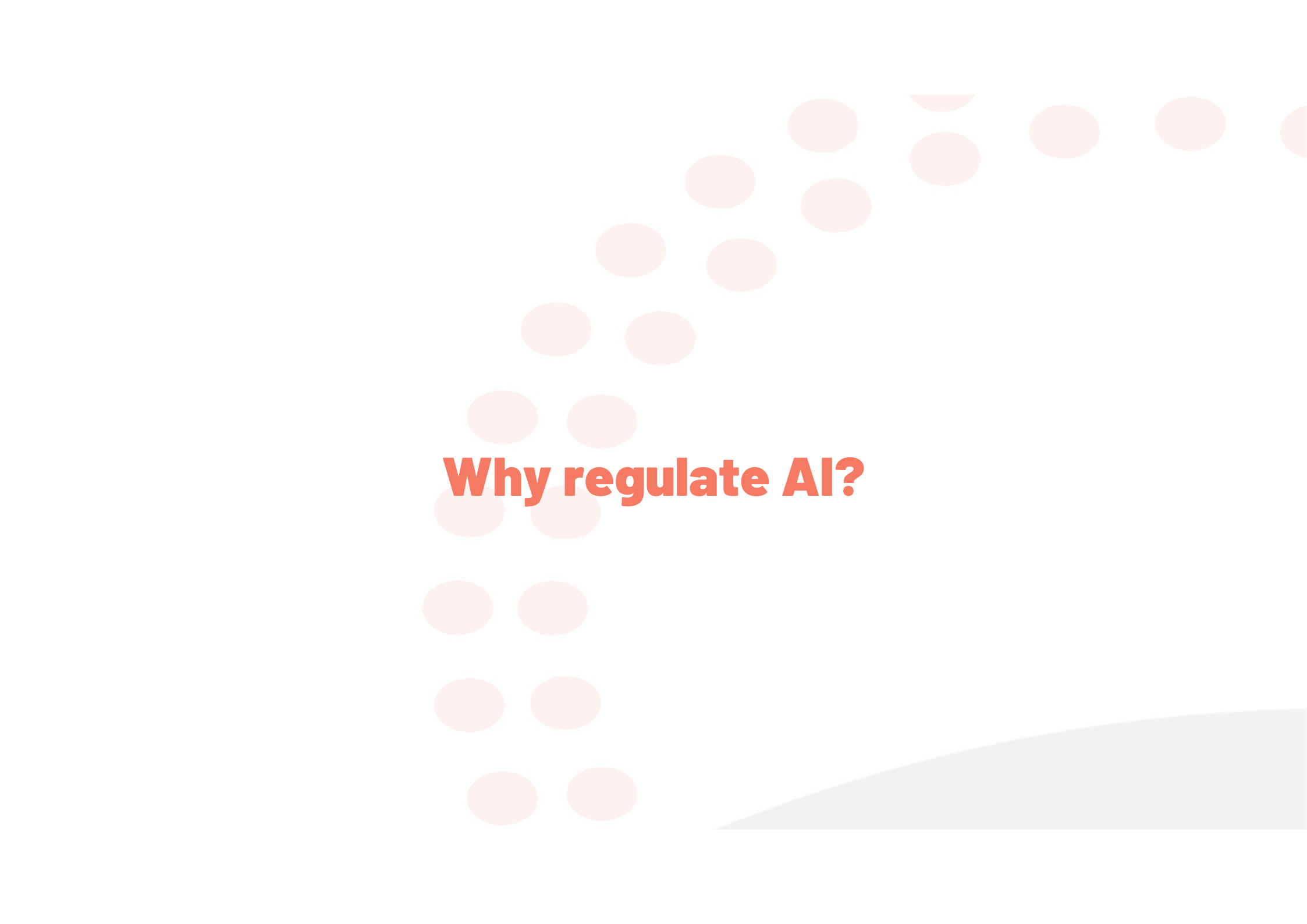
- **Monopolies** and natural monopolies
- Windfall profits ('economic rent')
- **Externalities** ('spillovers')
- **Information inadequacies**
- Continuity and availability of service
- Anti-competitive behaviour and predatory pricing
- Public goods and moral hazard
- **Unequal bargaining power**
- Scarcity and rationing
- Rationalisation and coordination
- **Planning** (e.g., for future generations)

Rights-based and social rationales (political)

- **Human rights**
- **Social objectives** (social solidarity)
- *E.g., regulating for reasons of distributional justice, rights protection, and citizenship*

Procedural approach (political)

- **Public dialogue and participation**
- Inclusive decision-making
- Deliberative democracy
- Transparency and accountability

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Why regulate AI?

Why regulate AI?

Public interest (welfare economic approach/market failure rationales)

- **Monopolies:** stifle competition and innovation, leading to higher prices and restricted access to AI applications. Example: AI infrastructure market: Amazon, Google, and Microsoft (Cloud computing, Usable data and Algorithms)
- **Information asymmetry:** recommendation systems/automated decision-making systems, addictive designs
- **Externalities:** e.g., environmental impact, surveillance (broader social harms) and racial and economic inequality (data labellers – the hidden workforce)

Why regulate AI?

Public interest (rights based and social rationales)

- **Injustice:** AI can create inefficiencies or injustices if algorithms are biased or not optimised for fairness. AI-powered hiring systems, trained on biased data, could continue to favour certain demographics (e.g., men over women in leadership roles), without regulation this can perpetuate gender inequality and harm society
- **Redistribution of resources:** regulation can be a tool for addressing inequalities by redistributing resources or opportunities in favour of disadvantaged groups (e.g., SMEs in AI industry, healthcare access).
- **Paternalism:** Regulation is used to protect individuals from making decisions that could harm their long-term interests or well-being (e.g., AI's role in decision-making; addictive design)
- **Shaping collective desires and preferences:** Regulation can reflect collective judgments about what society values, such as the importance of preserving cultural heritage or protecting human fundamental rights (privacy and dignity).
- **Further harms:** ensuring that AI technologies do not harm future generations by promoting sustainability and ethical innovation

GDPR

Public interest theories

- Protection of fundamental rights – legal objectives
- Information asymmetry: transparency and accountability
- Lawful processing → autonomy
- Art. 22 GDPR – Bias and discrimination

Art. 1 GDPR

Subject-matter and objectives

1. This Regulation lays down rules relating to the protection of natural persons with regard to the processing of personal data and rules relating to the free movement of personal data.
2. This Regulation protects fundamental rights and freedoms of natural persons and in particular their right to the protection of personal data.
3. The free movement of personal data within the Union shall be neither restricted nor prohibited for reasons connected with the protection of natural persons with regard to the processing of personal data.

AI Act

Public interest theories

- Improve the functioning of internal market
- Human-centric and trustworthy AI
- Protection of fundamental rights in EU charter
- Supporting innovation

Article 1: Subject Matter

Date of entry into force:

2 February 2025

According to:

Article 113(a)

Inherited from:

Chapter I

[See here for a full implementation timeline.](#)

SUMMARY +

1. The purpose of this Regulation is to improve the functioning of the internal market and promote the uptake of human-centric and trustworthy artificial intelligence (AI), while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy, the rule of law and environmental protection, against the harmful effects of AI systems in the Union and supporting innovation.

Theories of regulation

Interest group theories/Private interest theories

- Regulatory capture theory (rent-seeking)
- Public choice theory

Interest groups/Private interest theory and AI regulation

- Challenge the notion that regulation is always in the public's best interest: instead, these theories argue that regulation **often serves the interests of powerful groups**, such as businesses, unions, or political elites → who can influence the regulatory process to benefit themselves
- Regulatory decisions are shaped by **lobbying, economic power, and political influence** rather than solely addressing market failures or public good
- Regulatory environments as favours: these theories often argue that rather than correcting market failures, regulation is a tool for securing advantages for certain stakeholders (e.g., **subsidies, tax breaks, monopolistic protections**), often distorting market outcomes and benefiting the few rather than the many

Interest groups/Private interest theory and AI regulation

- E.g., Lobbying by tech giants: large tech companies like Google and Facebook have significant lobbying power

ANALYSIS

Big Tech Cozies Up to New Administration After Spending Record Sums on Lobbying Last Year

Tech giants combined to spend \$61.5 million on lobbying in 2024 — and employed one lobbyist for every two members of Congress

JAN 22, 2025

Theories of regulation

Institutionalist theories ('institutionalists') → institutional structure and arrangements
+ social processes significantly shape regulation

- Focus on the interdependence between state and non-state actors in the regulatory process, including governments, private companies, and civil society in shaping regulatory decisions.
- Bridge the gap between public and private interest theories
 - **Network theories and regulatory space:** regulatory networks → '**Smart regulation**' (e.g., mix of command-and-control methods with other regulatory instruments) → '**decentred regulation**' → regulatory authority is shared between private and public actors
 - Emphasise that regulatory strategies can be hybrid

Regulatory strategies/models

- **Command and control**

- Force of the law used to **prohibit certain forms of conduct**, to **demand positive actions** or to lay down conditions for entry into a sector (usually backed up by sanctions)
- **Risk of capture**
- **Risk of over-regulation**, legalism, delay intrusion or managerial freedoms & strangling of competition and enterprise (e.g., barrier to entry)
- **Enforcement is expensive** + effective of enforcement are uncertain

- **Self-regulation**

- A group of firms or individuals exerts control over its own membership and their behaviour
- It may operate in an informal, non-binding, voluntary manner, or it may involve rules of full legal force that are enforceable in the courts.
- Why self-regulate? **Expertise, control, legitimacy** → regulatory effectiveness
- However, risk of **lack of accountability** and **unfairness of procedures**

Regulatory strategies/models

- **Co-regulation, enforced self-regulation and meta-regulation**
 - **Co-regulation:** industry-association self-regulation with some oversight and/or ratification by government (Ayres and Brathwaite)
 - **Enforced self-regulation:** subcontracting (delegation) of regulatory functions to regulated firms (Ayres and Brathwaite)
 - **Meta-regulation:** processes in which the regulatory authority oversees a control or risk management system, rather than carries out regulation directly (Christina Parker) → **regulator's roles is auditing, monitoring and incentivising these systems**
 - **Cheaper** and **more effective** than command and control regimes
 - **Law and regulation** must hold corporate self-regulation accountable.
 - Liability or Accountability approach through courts or regulators
 - **Not problem-free:** Ill-intentioned, ill-informed, or inefficient firms may fail to devise appropriate rules → risk that regulatory processes will lack legitimacy and prove to be unfair, exclusive and inefficient.

Institutionalist theories and AI regulation

- **Co-regulation (institutional design):** involves cooperation between state and non-state actors, such as private companies and industry bodies, to set standards and regulations for AI
- **Dynamic governance:** recognises regulation of AI is not static; it involves continuous interaction among stakeholders as technologies and societal concerns change
- **Multi-level regulation:** AI regulation occurs at multiple levels (local, national, international), with each level influencing and being influenced by others.

Institutionalist theories and AI regulation

DSA as a co-regulation system

20 The DSA as a Co-regulatory System

[Get access >](#)

Martin Husovec

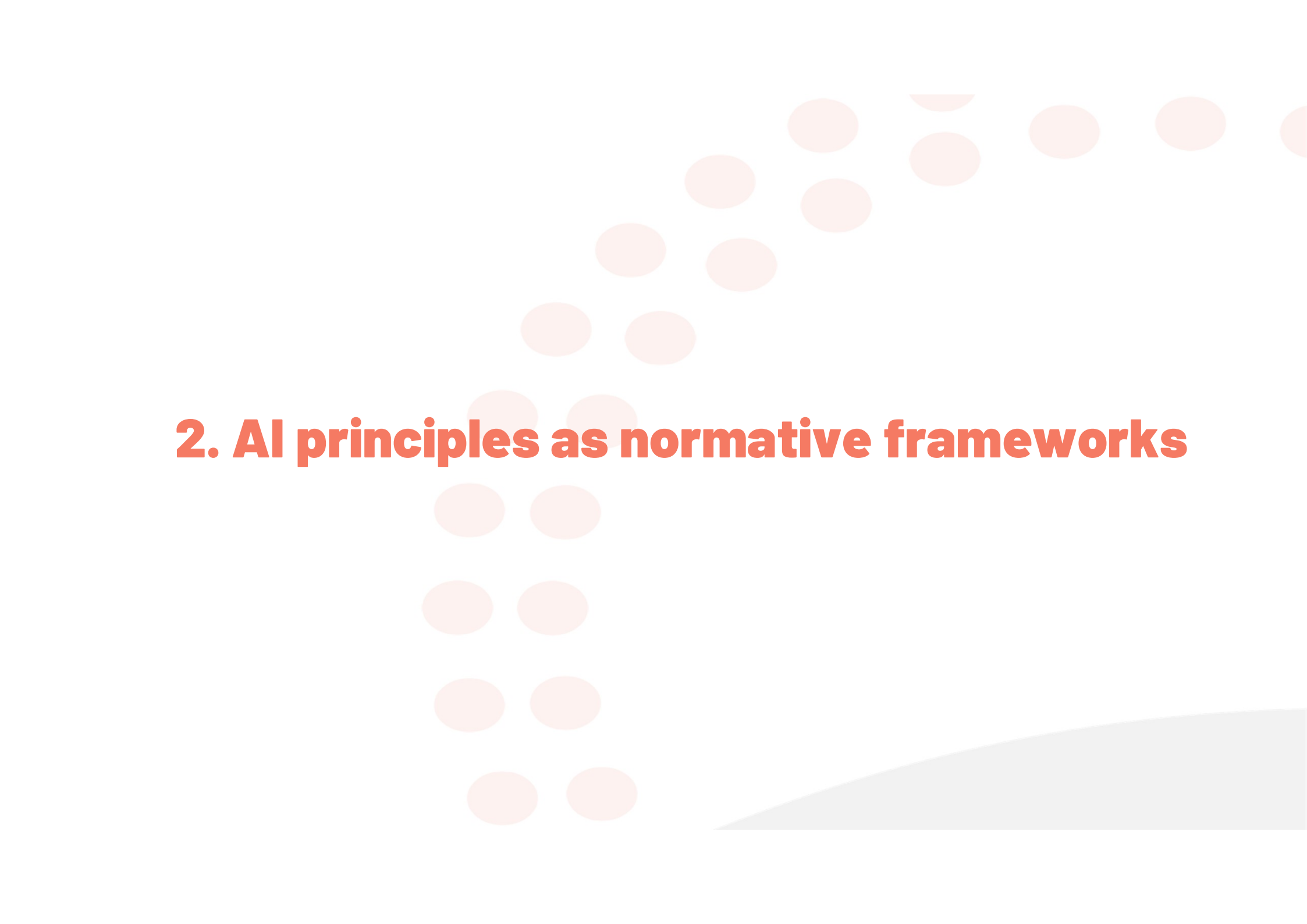
<https://doi.org/10.1093/law-ocl/9780192882455.003.0020> Pages 443–462

Published: August 2024

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Abstract

The chapter puts the newly introduced provisions of the Digital Services Act (DSA) into the broader regulatory landscape, arguing that one of the key components of the new law resides in co-regulation through Codes of Conduct. This approach has its roots in how the internet has been regulated historically. The chapter then conceptualises Codes of Conduct in terms of their future role in the enforcement of the DSA. After reviewing the different types of Codes of Conduct that the DSA envisages, it is argued that they are likely to have a crucial role in making the enforcement cheaper for the European Commission. Thus, a theory of effective enforcement is offered that rests on a tit-for-tat strategy where prescriptive rules and Codes of Conduct allow the Commission to punish companies to secure long-term cooperation on more open-ended rules, such as risk management.



2. AI principles as normative frameworks

PRINCIPLED ARTIFICIAL INTELLIGENCE

A Map of Ethical and Rights-Based Approaches to Principles for AI

DOCUMENT TIMELINE



Nature of Actors

- Civil Society
- Government
- Inter-governmental Organization
- Multistakeholder
- Private Sector

PRINCIPLED ARTIFICIAL INTELLIGENCE

A Map of Ethical and Rights-Based Approaches to Principles for AI

Authors: Jessica Fjeld, Nele Achten, Hannah Hilligoss, Adam Nagy, Madhulika Srikumar

Designers: Arushi Singh (arushisingh.net) and Melissa Axelrod (melissaaxelrod.com)

HOW TO READ:

Date, Location
Document Title
Actor

COVERAGE OF THEMES:



The size of each dot represents the percentage of principles in that theme contained in the document. Since the number of principles per theme varies, it's informative to compare dot sizes within a theme but not between themes.

The principles within each theme are:

Privacy:

Privacy
Control over Use of Data
Consent
Privacy by Design
Recommendation for Data Protection Laws
Ability to Restrict Processing
Right to Rectification
Right to Erasure

Accountability:

Accountability
Recommendation for New Regulations
Impact Assessment
Evaluation and Auditing Requirement
Verifiability and Replicability
Liability and Legal Responsibility
Ability to Appeal
Environmental Responsibility
Creation of a Monitoring Body
Remedy for Automated Decision

Safety and Security:

Security
Safety and Reliability
Predictability
Security by Design

Transparency and Explainability:

Explainability
Transparency
Open Source Data and Algorithms
Notification when Interacting with an AI
Notification when AI Makes a Decision about an Individual
Regular Reporting Requirement
Right to Information
Open Procurement (for Government)

Fairness and Non-discrimination:

Non-discrimination and the Prevention of Bias
Fairness
Inclusiveness in Design
Inclusiveness in Impact
Representative and High Quality Data
Equality

Human Control of Technology:

Human Control of Technology
Human Review of Automated Decision
Ability to Opt out of Automated Decision

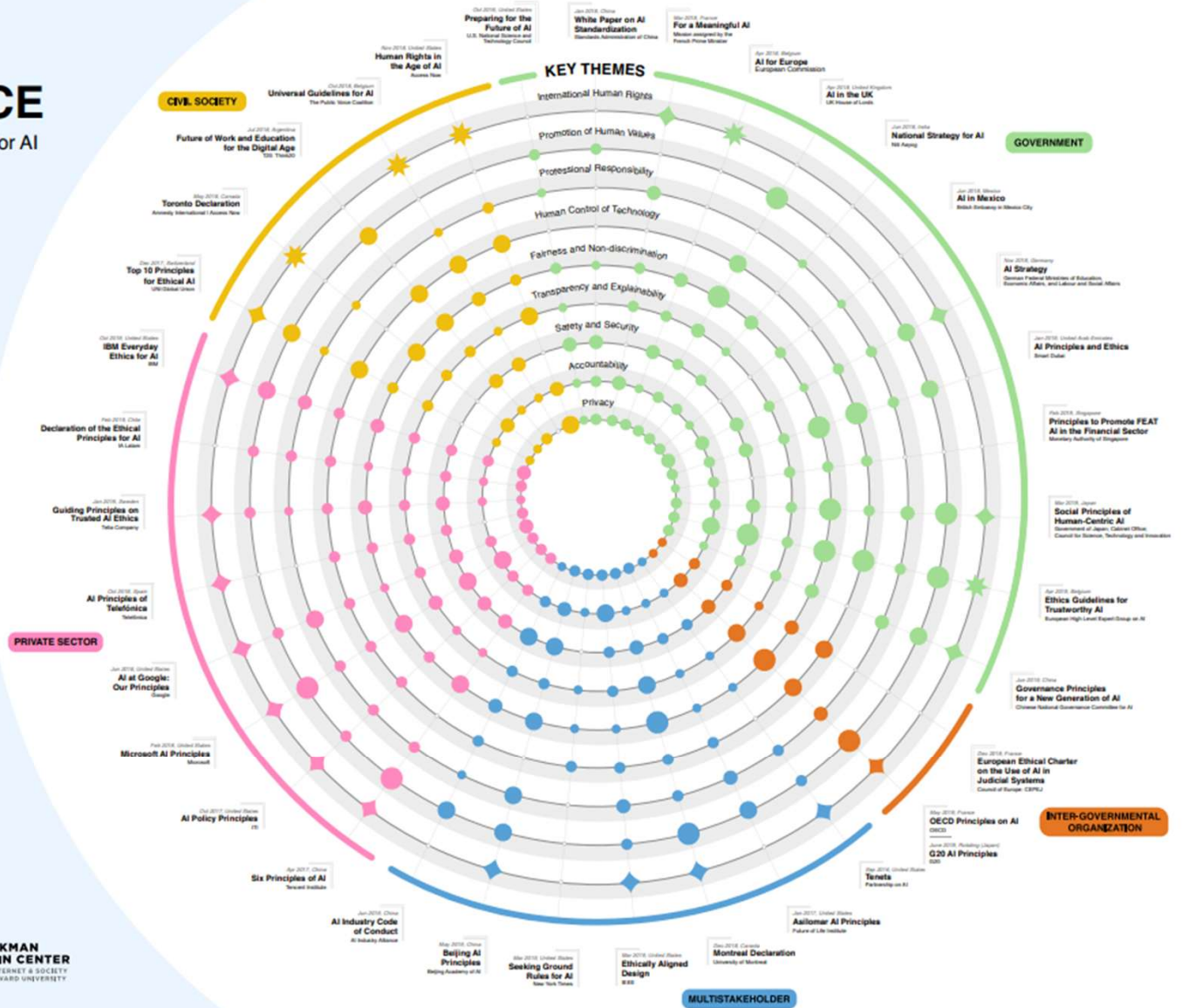
Professional Responsibility:

Multistakeholder Collaboration
Responsible Design
Consideration of Long Term Effects
Accuracy
Scientific Integrity

Promotion of Human Values:

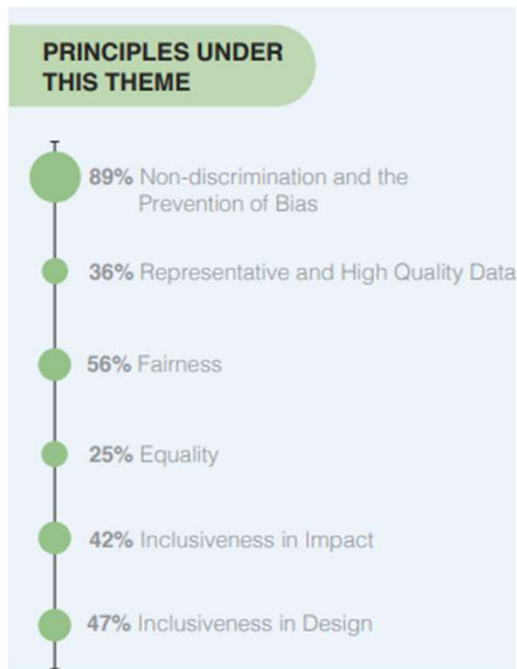
Leveraged to Benefit Society
Human Values and Human Flourishing
Access to Technology

BERKMAN
KLEIN CENTER
FOR INTERNET & SOCIETY
AT HARVARD UNIVERSITY

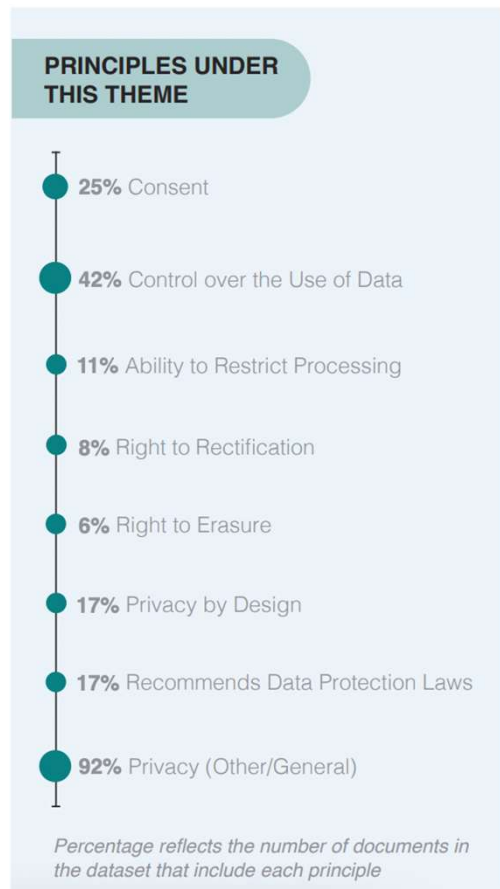


Ethical principles for AI

Fairness & Non-discrimination (100%)



Privacy (97%)



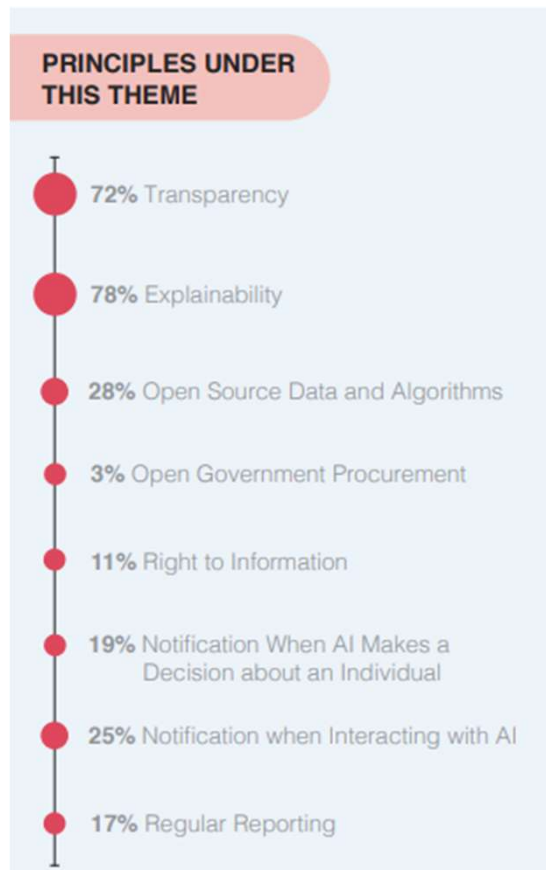
Accountability (97%)



From Fjeld, J. et al. (2020). Principled artificial intelligence: Mapping consensus in ethical and rights-based approaches to principles for AI. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3518482

Ethical principles for AI

Transparency & Explainability (94%)



Safe & Security (81%)



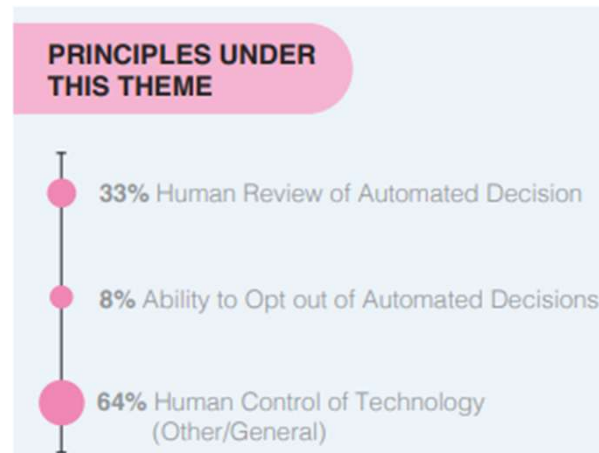
Professional responsibility (78%)



From Fjeld, J. et al. (2020). Principled artificial intelligence: Mapping consensus in ethical and rights-based approaches to principles for AI. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3518482

Ethical principles for AI

Human Control of Technology (69%)



Promotion of Human Values (69%)



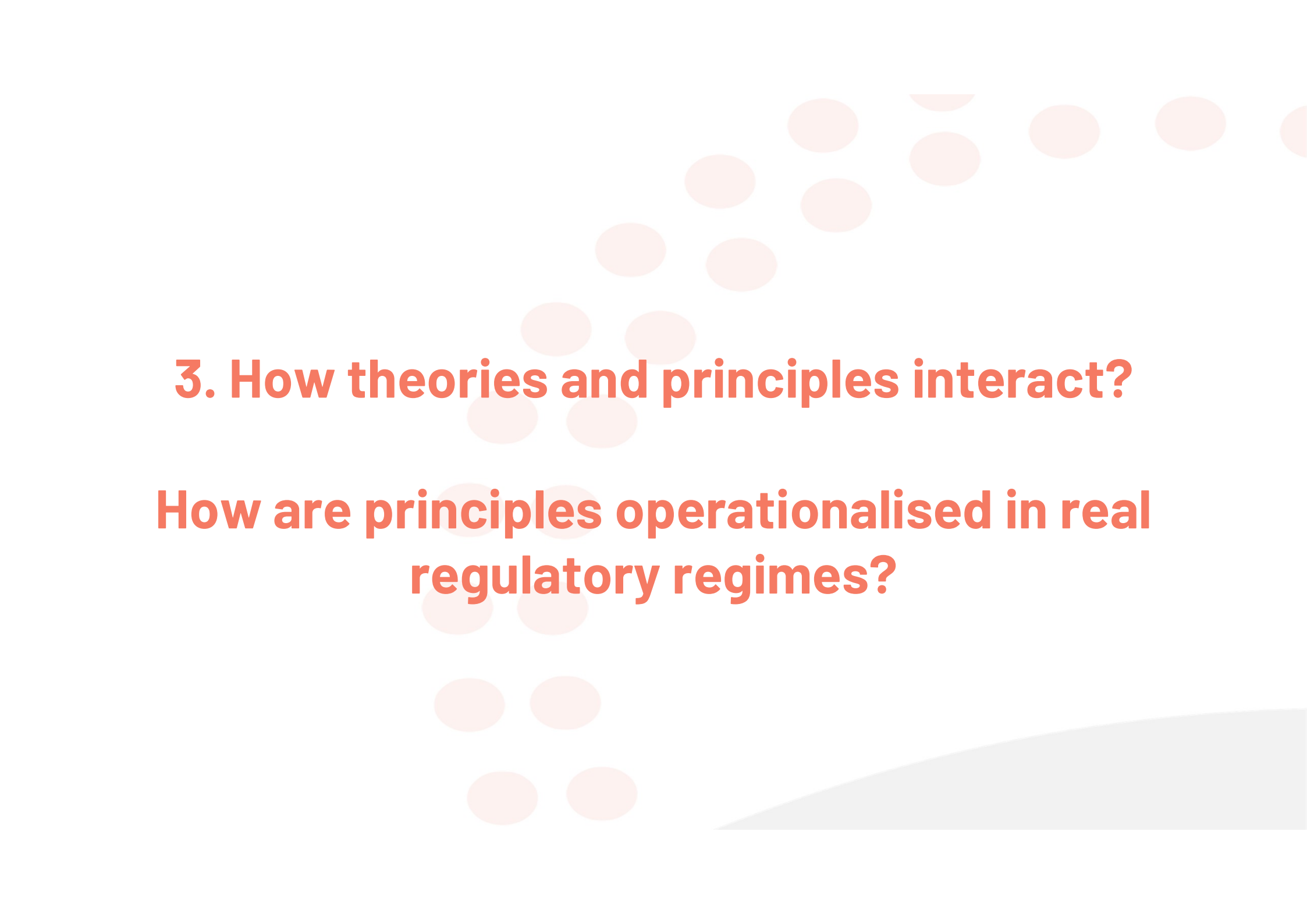
From Fjeld, J. et al. (2020). Principled artificial intelligence: Mapping consensus in ethical and rights-based approaches to principles for AI. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3518482

Group discussion (10 minutes)

- ***There are two main perspectives on how AI regulation should be structured. On one side, some argue for a polycentric (multistakeholder) regulatory system, where multiple actors – including governments, private sector companies, NGOs and other stakeholders – collaborate to regulate AI systems. On the other side, others advocate for consolidated oversight, where AI regulation is managed by a single central body to ensure consistency and efficiency.***
- ***Which model do you recommend and for which types of AI systems? Please explain your reasons?***



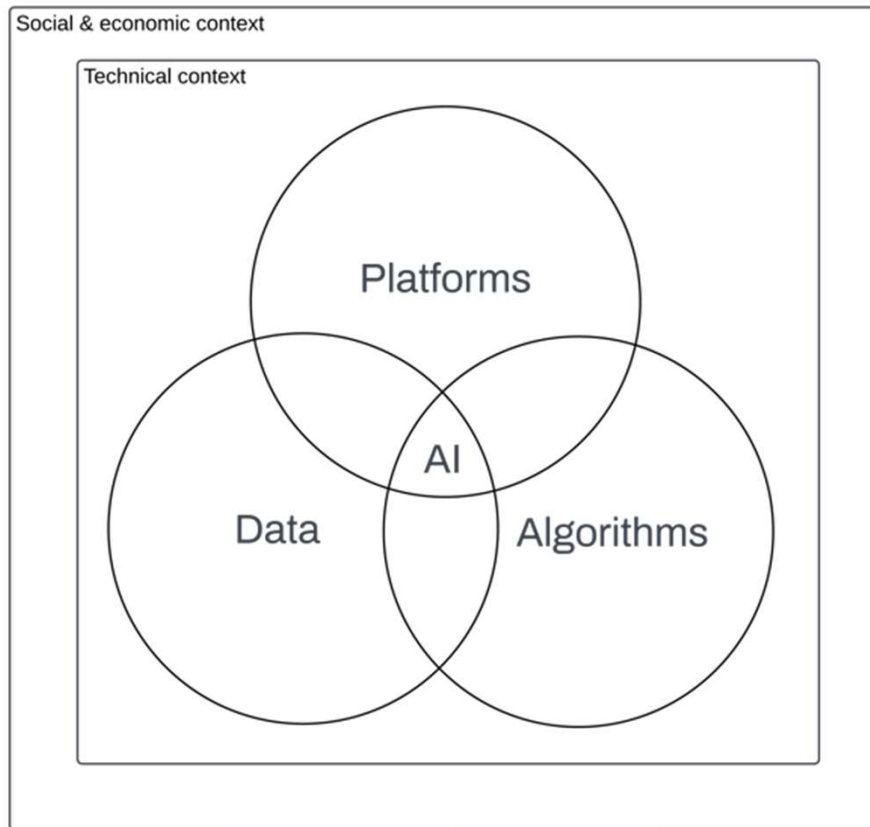
What do you think are the biggest ethical and legal challenges in governing AI?



3. How theories and principles interact?

How are principles operationalised in real regulatory regimes?

Governing the 'technology toolbox'



- Nitzberg & Zysman suggest that governing the full “technology toolbox” is critical for navigating the tensions inherent to AI governance.
- The toolbox consists of algorithms, data, and platforms that operate within specific technical, social, & economic contexts

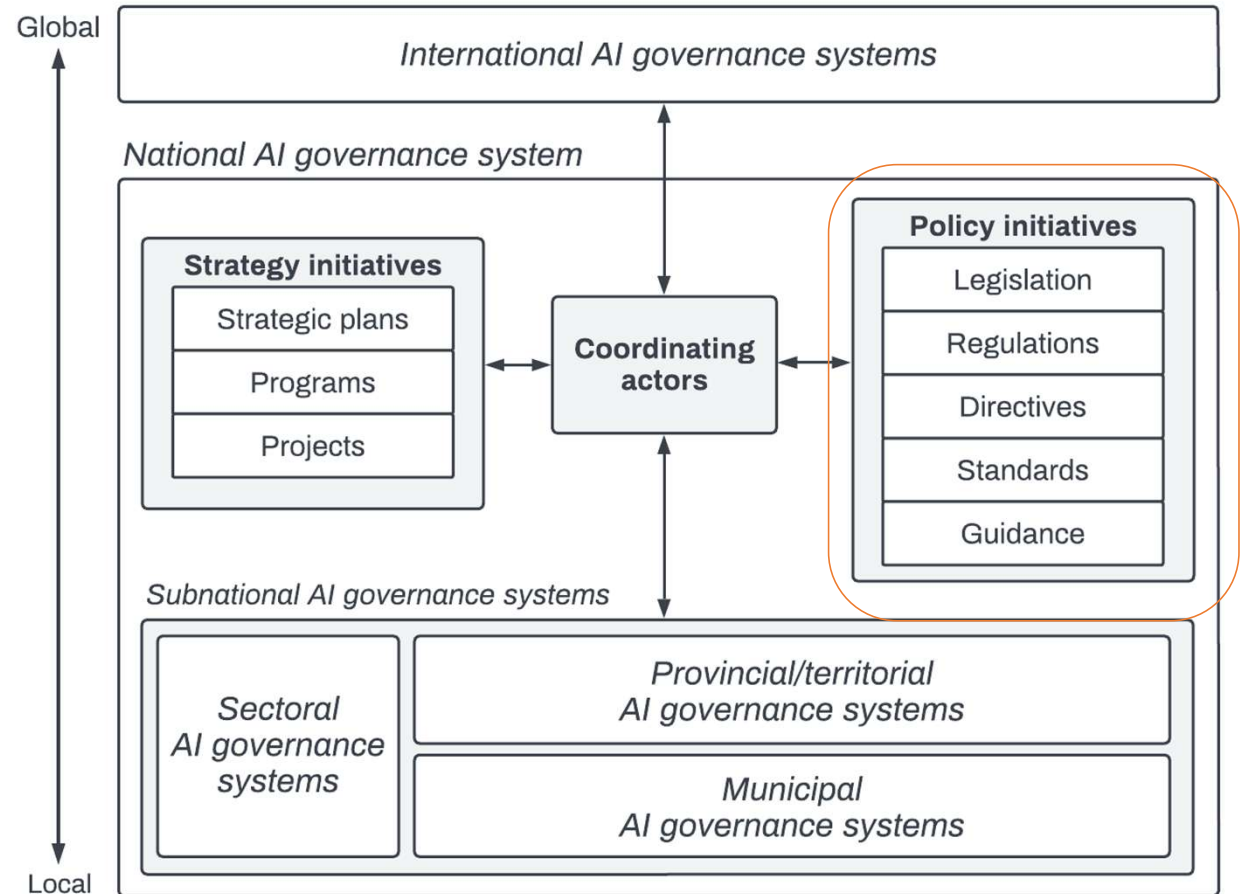
Diagram based on Nitzberg, M. & Zysman, J. (2022). Algorithms, data, and platforms: The diverse challenges of governing AI. *Journal of European Public Policy*, 29(11), 1753-1778. <https://www.tandfonline.com/doi/pdf/10.1080/13501763.2022.2096668>

Different regulatory approaches to regulate AI

- **Risk-based approach** focuses on the control of risks associated with certain types of algorithmic applications from a public-interest perspective – instead of the compliance with rules. The approach prioritises objectives, differentiates risk types, and selects interventions based on risk levels.
- **Human-rights-based approach** aims at ensuring that AI regulations protect individuals' rights and freedoms, emphasising human rights. This approach assumes that market failures are not the only rationales for regulating but that regulations are justified to protect rights, promote distributional justice, and further social objectives
- **Principles-based approach** offers a set of fundamental propositions (principles) that provide guidance for developing and using AI systems through ethical, responsible, human-centric, and human-rights-abiding processes.
- **Agile and experimentalist approach:** focuses on adopting agile approaches to generate flexible regulatory schemes, such as regulatory sandboxes and testbeds, in diverse economic sectors.
- **Adapting existing law approach:** adapts sector-specific rules (e.g., health, finance, education, justice) and transversal rules (e.g., criminal codes, public procurement, data protection laws, labour laws) instead of issuing AI bills.
- **Liability approach:** assigns responsibility and sanctions to problematic use of AI systems

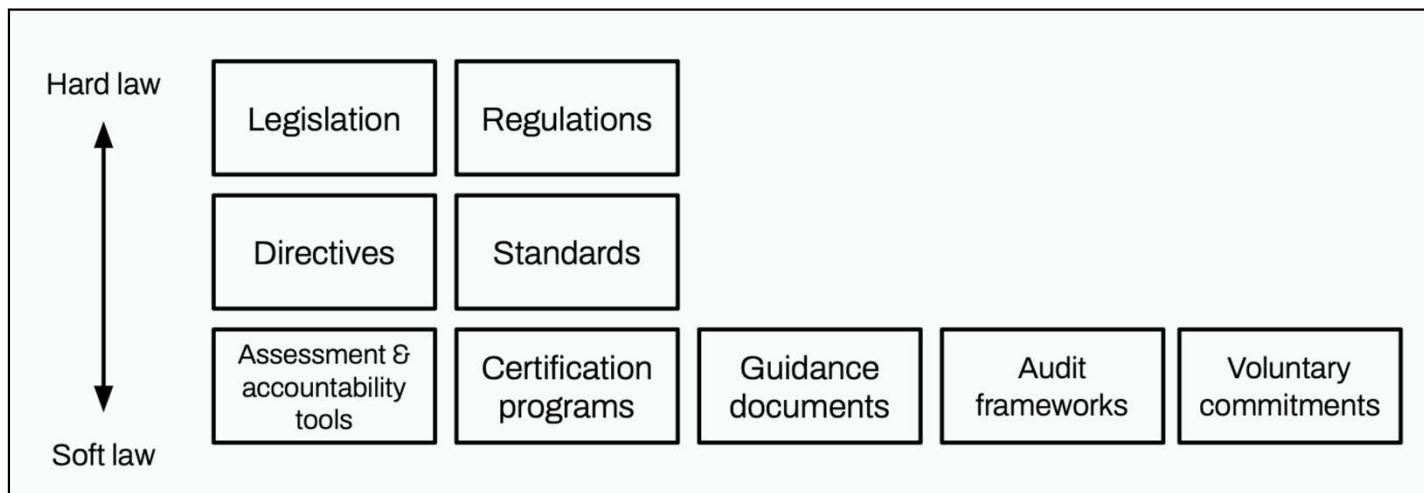
Global 'trends' for AI regulation

– Policy tools and instruments



Types of policy instruments: Hard law v. Soft law

- **Hard law** instruments require mandatory compliance and are made legally binding by legislation and regulations
- **Soft law** instruments encourage voluntary compliance and are typically targeted toward managers & practitioners
- Because AI policy is still an emerging space, it is heavily reliant on soft law instruments as legislation & regulation continue to develop



EU AI Act – Risk-based approach to AI

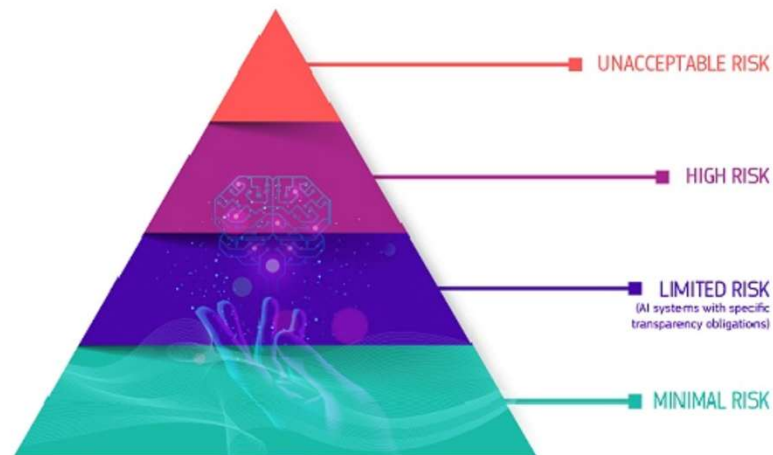
Risk-based Approach

Compliance and enforcement

Simplification Proposal

A Risk-based Approach

The AI Act defines 4 levels of risk for AI systems:



Unacceptable risk

All AI systems considered a clear threat to the safety, livelihoods and rights of people are banned.

UNESCO's Recommendations on the Ethics of AI - Principles-based approach to AI (Standards-based)



UK pro-innovation approach to AI Regulation



Department for
Science,
Innovation
& Technology



Office for Artificial
Intelligence

Policy paper

A pro-innovation approach to AI regulation

Updated 3 August 2023

10. Our framework is underpinned by 5 principles to guide and inform the responsible development and use of AI in all sectors of the economy:

- Safety, security and robustness
- Appropriate transparency and explainability
- Fairness
- Accountability and governance
- Contestability and redress

11. We will not put these principles on a statutory footing initially. New rigid and onerous legislative requirements on businesses could hold back AI innovation and reduce our ability to respond quickly and in a proportionate way to future technological advances. Instead, the principles will be issued on a non-statutory basis and implemented by existing regulators. This approach makes use of regulators' domain-specific expertise to tailor the implementation of the principles to the specific context in which AI is used. During the initial period of implementation, we will continue to collaborate with regulators to identify any barriers to the proportionate application of the principles, and evaluate whether the non-statutory framework is having the desired effect.

From UK Office for Artificial Intelligence (2023). A pro-innovation approach to AI regulation.
<https://www.gov.uk/government/publications/ai-regulation-a-pro-innovation-approach/white-paper>

UK pro-innovation approach to AI Regulation

14. We have identified a number of central support functions required to make sure that the overall framework offers a proportionate but effective response to risk while promoting innovation across the regulatory landscape:

- Monitoring and evaluation of the overall regulatory framework's effectiveness and the implementation of the principles, including the extent to which implementation supports innovation. This will allow us to remain responsive and adapt the framework if necessary, including where it needs to be adapted to remain effective in the context of developments in AI's capabilities and the state of the art.
- Assessing and monitoring risks across the economy arising from AI.
- Conducting horizon scanning and gap analysis, including by convening industry, to inform a coherent response to emerging AI technology trends.
- Supporting testbeds and sandbox initiatives to help AI innovators get new technologies to market.
- Providing education and awareness to give clarity to businesses and empower citizens to make their voices heard as part of the ongoing iteration of the framework.
- Promoting interoperability with international regulatory frameworks.

But.. What about AI regulation under the new UK government?

From UK Office for Artificial Intelligence (2023). A pro-innovation approach to AI regulation.
<https://www.gov.uk/government/publications/ai-regulation-a-pro-innovation-approach/white-paper>

Deregulation

Reducing and eliminating government regulations

Reducing those that are perceived as overly burdensome or restrictive. The aim is to create a more favourable environment for innovation and market competition by allowing industries, including AI, to grow with minimal state intervention.

Global Competitiveness

Countries and regions that impose stringent AI regulations may risk falling behind competitors that offer looser regulatory environments. In the race for global AI leadership, governments may reduce or delay regulations to allow AI companies to develop and deploy their technologies faster.

- The EU AI Act
- GDPR (General Data Protection Regulation) → **competitive disadvantage for Europe? (regulatory convergence)**

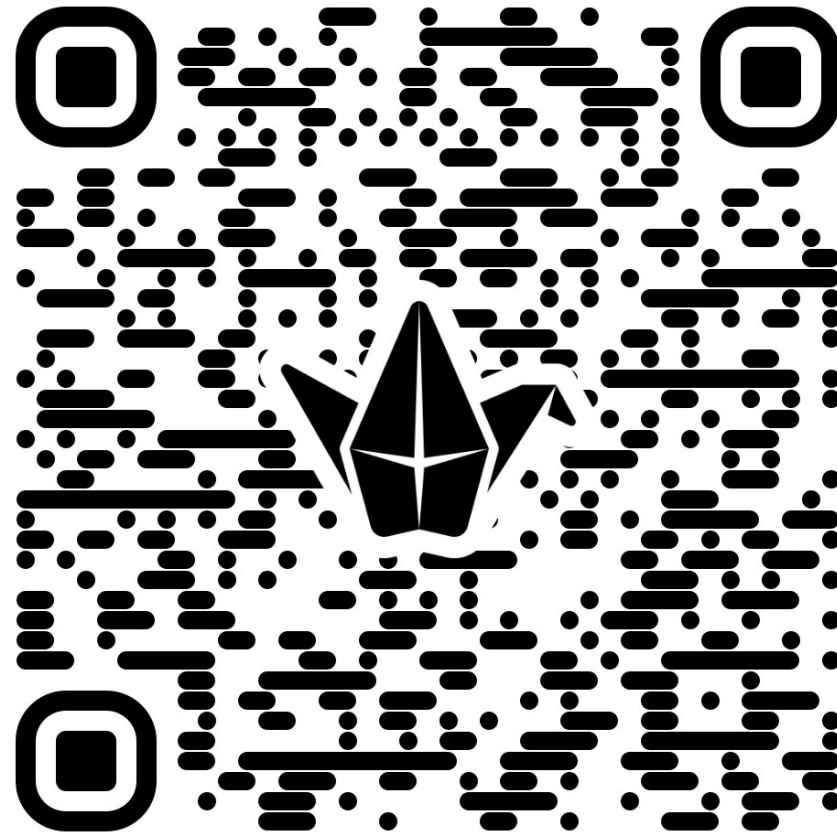
Reducing Bureaucratic Hurdles

Excessive regulation hinders growth and delays the time-to-market for AI products. Deregulation is seen as a means to remove red tape, making it easier for small companies and new innovators to enter the AI market without the burden of stringent compliance costs.



4. Group activity

Group Activity



<https://padlet.com/alineiramina/my-stunning-sandbox-xdltilvf8r6u8b7h>

From principle to regulation design activity

- 1. Each group now select one of the eight themes (principles)**
- 2. You are a policymaker and your task is to design a regulation that operationalises this theme and one of its principles in a real-world context**
- 3. Groups should answer five questions (write in the padlet):**
 - **Why regulate?**
 - **Which regulatory tool(s) would you choose?**
 - **What would compliance look like?**
 - **What is the regulatory approach and why?**
- 4. 'Regulatory gap analysis'**
 - **Conflicts with any other principles?**
 - **Unintended consequences (based e.g., regulatory theory)**

Thank you!